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Magnetostratigraphy of the Shungura and Usno Formations, southwestern Ethiopia: new data and comprehensive reanalysis

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Summary. Six hundred new samples are added to those we previously reported from the Shungura and Usno Formations of southwestern Ethiopia, and the entire data set is reviewed. Limited mineralogical study of sands indicates maghemite and hematite as carriers of remanence. After AF cleaning the remanent directions are well polarized into normal and reversed groups, with greater scatter in reversed directions. The time-averaged magnetization axis corresponds to a 'far-sided right-hand' virtual pole, a pattern previously found by Wilson. Intrasite dispersion is commonly large but better for the new samples, which we attribute to field technique. Polarity zonation is accomplished by statistical smoothing on stratigraphic plots of angle from dipole axis. Magnetozones are correlated with the standard polarity-chronologic intervals using K–A dates at nine stratigraphic levels. The main change from our previous interpretation is that the reversed zone in the Basal Member of the Shungura Formation is correlated with the top of the Gilbert Epoch rather than with the Mammoth event of the Gauss Epoch. Thus the Basal Member and Member A are somewhat older than we previously believed, but age assignments for major fossil localities are unaffected. The inferred age ranges for the Shungura and Usno Formations are 3.5 to 0.8 Myr and 3.7 to 2.8 Myr respectively. The present analysis yields five zones of normal polarity in the lower Matuyama Reversed Epoch, that is, between the Matuyama/Gauss transition and the base of the Olduvai. All were found both in the previous data and in the new samples taken at different locations. For three of the five zones an unbroken series of normal samples extends over several lithologies, so a geomagnetic event is indicated.

Introduction

The Shungura and Usno Formations of southwestern Ethiopia (Fig. 1) have yielded a wealth of vertebrate fossils (> 30 000 specimens), including fossils of early hominids. These formations have received intensive geological and palaeontological study over the past decade (Coppens *et al.* 1976). Previously we published on the magnetostratigraphy of the Shungura Formation (Shuey, Brown & Croes 1974), and subsequently magnetic data were added for

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the nearby Usno Formation (Brown & Shuey 1976). Both reports were based on samples collected in 1973 July and August. For some parts of the section the polarity was poorly defined so that polarity-zone assignments were rather tenuous. In view of the importance of these strata we wanted to improve the reliability of their magnetostratigraphy. To this end an additional 600 samples were taken in 1974 from 174 stratigraphic levels of the Shungura Formation. The present reanalysis is based on all measurements of these samples and the earlier samples taken in 1973.

Lithology and lithostratigraphy

Lithologies of the Pliocene/Pleistocene Shungura and Usno Formations range from coarse sand to clay and tuff. Major tuffs have enabled division of the Shungura Formation into members lettered A to L, and a Basal Member below Tuff A (de Heinzelin, Brown & Howell 1970). Locally both formations have been subdivided into lithostratigraphic units labelled numerically (de Heinzelin *et al.* 1970). The tuffs consist primarily of volcanic glass, and appear little altered. While a few of the tuffs have massive bedding, most have been reworked by water and have crossbeds, ripple marks and other sedimentary structures. Sands are most often found at the base of a submember. They are often crossbedded, but sometimes have planar bedding, or are massive. Some are partially cemented by calcium carbonate, and some have small concretions cemented by hydrous iron oxides. Occasionally large tubular concretions interpreted as rootcasts are observed.

The finer sediments (silts and clays) generally comprise the upper parts of submembers, but in the interval from G-14 to G-28 they are the dominant lithology. The silts range from sandy to clayey varieties, and are massive or laminated. They often contain rootcasts or casts of rootlets made up of calcium carbonate, or nodular concretions of calcium carbonate. The clays and clayey silts often have slickensides, presumably formed as a result of wet–dry cycles when the sediments were exposed at the land surface. In the stratigraphic interval from G-14 to G-28 the silts, silty clays and clays are most often finely laminated, contain crystals and concretions of gypsum and sometimes halite. The dominant clay mineral throughout the section is montmorillonite, but kaolinite appears in some samples in smaller amount.

The sands are interpreted as fluvial channel deposits or proximal deltaic sands (in G-14 to G-28). The silts and clays are interpreted as levee and floodbasin deposits except in the interval G-14 to G-28 where they are regarded as being primarily of lacustrine origin.

Field work

As far as possible, the palaeomagnetic sampling was conducted along detailed profiles mapped topographically and geologically at a scale of 1:200 by de Heinzelin, Haesaerts & Howell (1976). These type sections are frequently close to major fossil localities. Fig. 1 locates the palaeomagnetic sampling sites for both the 1973 and 1974 collections. The strike of the beds is north and the dip is to the west (usually about 10°). The sampling procedure along a profile was to dig pits at stratigraphic intervals of 1–2 m and cut three or four cubes in each pit. One surface of a cube was coincident with the bedding plane, and one edge was oriented north by compass. During laboratory work on samples taken in 1973 it was noted that samples from nearly horizontal surfaces tended to yield poorer results than those taken from steep surfaces; this was attributed to weathering (Shuey *et al.* 1974). In order to minimize these effects samples were obtained in 1974 from steep outcrops wherever possible, and sampling pits were dug deeper than before. In addition, fine grained sediments (clays and silts) were sampled whenever possible rather than sands, because in a portion of

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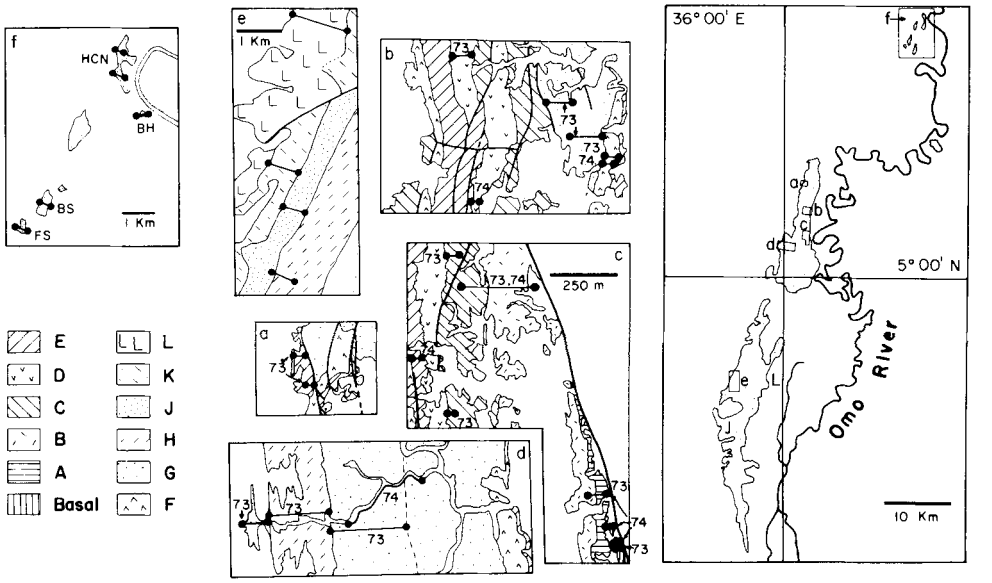


Figure 1. The large map shows the general distribution of the Shungura and Usno Formations, and the locations of detailed portions a, b, c, d, e and f. The detailed map (a) is a simplified geologic map with the sections marked indicating the area where Member E was sampled; (b) shows profiles along which Member D was sampled in 1973 and 1974, and along which Member F was sampled in 1973 and 1974; (c) shows profiles along which the Basal Member and Members A, B and C were sampled, and along which Member E was resampled in 1974; (d) indicates lines along which upper Member G, Member H and Member J were sampled in 1973, and the gully along which G-11 to G-25 were resampled in 1974; (e) shows lines along which upper Member H, and Members J, K and L were sampled in 1973 in the Kalam area; (f) shows lines along which the Usno Formation was sampled. The locality letters correspond to the letters alongside the stratigraphic column of Fig. 6. The scale marked for (c) is applicable to (a), (b) and (d). Scales for (e) and (f) are given. The letters J and L refer to 1974 sampling localities for those members respectively.

Table 1. Portions of the Shungura Formation resampled in 1974.

Stratigraphic level	Interpretive status from 1973 data
Member L (part)	Upper Matuyama epoch and Jaramillo event
Member J (part)	Upper Matuyama epoch long reversed interval
G-10 to G-24	Mixed polarity
G-2 to G-24	Mixed polarity, sparse samples
F-3 to F-5	'Reunion I' normal event
All Member E	Reversed, but poor quality
D-3 to D-5	'X' normal event
C-8 to C-9	Gauss/Matuyama epoch boundary
B-2 to B-3	Kaena reversed event
All Basal Member	Indeterminate polarity

the 1973 collection a correlation of normal polarity with sandstone lithology was observed and attributed to chemical remagnetization of the sands.

Some stratigraphic intervals were sampled at more than one geographic location (Fig. 1). This enabled us to test the lateral continuity of some magnetozones. The objectives of the 1974 resampling programme of the Shungura Formation (Table 1) were first to test the magnetozone boundaries inferred from the 1973 data and secondly to try to improve the worst parts of our data through the improved sampling procedure.

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Magnetic mineralogy

We separated a magnetic fraction from several resuspended sands using a hand magnet. On the basis of the experience reported by Løvlie, Lowrie & Jacobs (1971), we would expect this separation to be fairly efficient for spinel-type iron oxides (magnetite and maghemite) but very inefficient for hematite, hydrated iron oxides, or paramagnetic minerals. X-ray diffraction patterns of these separates showed definitely that the principal mineral was a spinel-type oxide, and a line at 3.75 Å implied the presence of maghemite. Microscopic examination of polished sections of this material confirmed the predominance of maghemite and revealed ubiquitous exsolution lamellae within the maghemite grains. Semi-quantitative analysis of these lamellae using a scanning electron microscope indicated iron and titanium in roughly equal proportions, so we regard the lamellae as ilmenite. The sections also contained a hydrated iron oxide, but magnetite was not seen.

These data suggest that in the sands maghemite and not magnetite is a significant contributor to the measured remanence. The source area for the sediments of the Shungura and Usno Formations is believed to lie in volcanic highlands to the north, so the maghemite is likely to be an oxidation product of volcanic magnetite. Did this oxidation occur in the source region, the depositional environment, or the contemporary weathering environment? Mineralogical data for the present soils and river sediments might be relevant to this question, but suitable samples are not now available to us. We did investigate one sand sample from the late Pleistocene Kibish Formation (de Heinzelin *et al.* 1970), which unconformably overlies the Shungura Formation and is thought to have a similar source area. The magnetic separate from the Kibish Formation was also dominated by exsolved maghemite, but in addition contained a small amount of magnetite. This datum suggests the hypothesis that the destruction of magnetite occurred in part after deposition of the Shungura and Usno Formations.

We performed stepwise thermal demagnetization at 50°C increments in air on a few whole samples using a Schoenstedt TSD-1 oven (heating field 150 gamma, cooling field 10 gamma). Distinct reddening indicated the formation of hematite. In addition, the intensity remaining at 600°C was surprisingly high, about 10 per cent of the initial intensity. The directional measurements from the 600–650°C range were close to the directional measurements from the 200–600°C range, regardless of orientation of the sample in the oven. The most likely explanation of this observation is that some of the NRM is due to hematite which is magnetized in the same direction as any other components.

From these data it is clear that we are not dealing with simple detrital magnetite but rather with oxidized sediments in which maghemite and hematite are contributing to remanence. For late Cenozoic red sediments in Baja, California, Larson & Walker (1975) reported that different component minerals (hematite, geothite, and 'hydropsilomelane') had different polarities so that where several components were present the net magnetization was in an intermediate direction (i.e. neither normal nor reversed). We do not have the same situation because for most samples a moderate partial demagnetization results in clear normal or reverse polarity and further demagnetization does not generally change the direction.

Laboratory procedure

Remanent magnetization was measured on a DIGICO spinner magnetometer using 32 to 128 revolutions about each spin axis, depending on magnetic intensity. Repeated six-spin measurements on a few samples gave an angular standard deviation of less than 2°. Immediately after the 1973 field season we carried one sample per site through progressive

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AF demagnetization measuring declination only (i.e. with two spins instead of six). This enabled us to define zones of consistent polarity and zones of difficulty, where declination was not consistent from one demagnetization step to the next, not consistent from one stratum to the next, or not close to either 0 or 180°. Next we made vector measurements (six-spins) on a second sample per site, again with progressive AF demagnetization. Where there was any inconsistency of polarity between the two samples at a site or between consecutive sites, we made vector measurements on all other available samples from the site and on the first sample which had been partially demagnetized with only declination monitored. The 1974 samples were taken only from zones of difficulty, and so all were progressively demagnetized with vector measurements.

A common procedure in routine AF demagnetization is to take a few pilot samples to a high AF field in small increments and then select a single optimum field strength for demagnetization of the bulk of the samples. In contrast we took all samples through small increments (25 or 50 gauss) until an optimum field strength was reached. Then a few samples were taken to higher fields (up to 700 G) to guard against misinterpretation of the sequence of vector measurements at low and intermediate fields (up to 250 G). Where there was inconsistency of polarity within a site or between consecutive sites, the relevant samples were demagnetized further.

We think it is appropriate to have chosen an optimum demagnetizing field individually for each sample because the samples differed greatly in the response of remanent vector to partial demagnetization. The several samples from a single site behaved about the same upon demagnetization, but stratigraphically consecutive sites frequently had different magnetic behaviour.

In choosing one field as optimum, we looked for four things: (1) direction within 40° of the axial dipole field; (2) reduction of intensity to 10–50 per cent of the initial value; (3) consistency of direction in consecutive demagnetization steps, or else regular movement towards a determinate direction. If all three conditions were met at some field we tentatively accepted that as optimum. If there was inconsistency of polarity within a site or between consecutive sites, we would continue demagnetization to higher fields until reaching a minimum of intensity and/or non-repeatability of measurements at a single peak field. This limit is evidently set by ambient laboratory fields and generally ranged from 400 to 700 G. The optimum field was then taken as the one preceding the terminal field just described. In no case did we choose an optimum field for a site by the criterion of minimum intrasite angular dispersion, as is sometimes done, although we do think such a procedure would be a valid alternative to our actual procedure.

Directional statistics

The AF partial demagnetization resolved the samples into two polarities, Fig. 2 shows histograms of declination for NRM and after demagnetization at the field selected for each sample as just described. Note that partial demagnetization decreases the number of samples at intermediate declinations and increases the number of samples at reversed declinations leaving the peak at normal declinations relatively unchanged. The reversed peak is somewhat broader than the normal peak; the standard deviations are about 20 and 17° respectively.

Fig. 3 is a stereographic (equal-area) plot of the cleaned directions. The number of samples is somewhat less than in Fig. 2 because we did not make vector measurements on the initially measured samples where there was no inconsistency of polarity within a site and between consecutive sites. We do not think anything significant is to be gained by additional work, since our magnetic zonation (below) would be the same even omitting those samples

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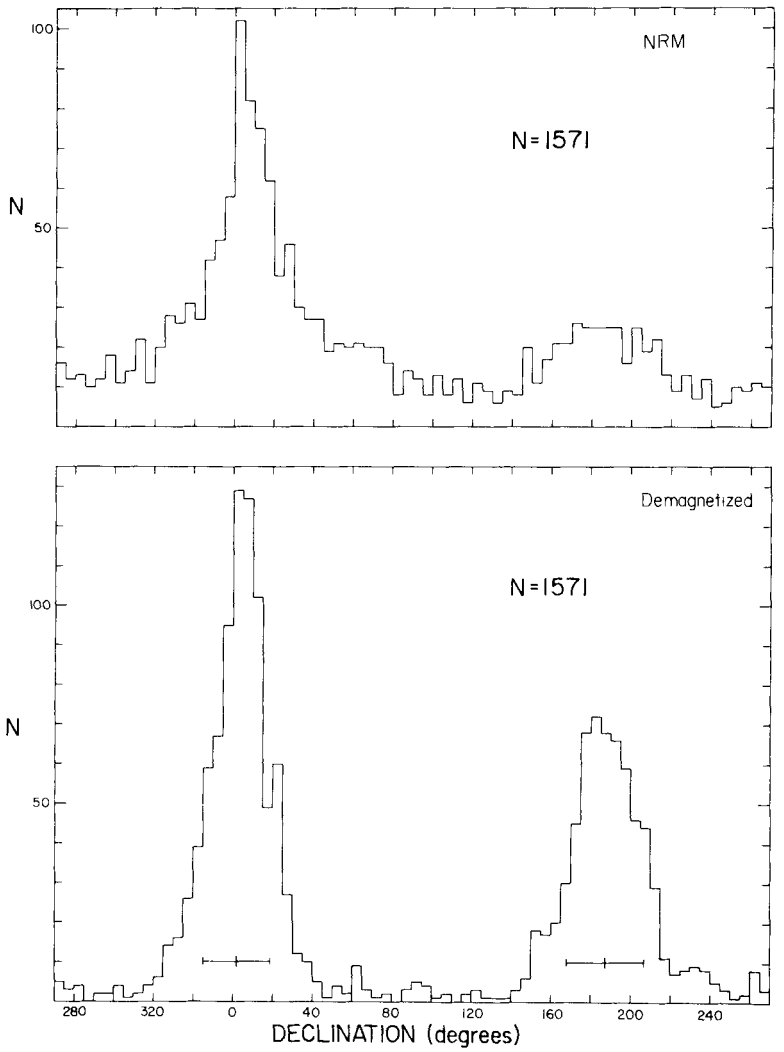


Figure 2. Histograms of remanent declinations in all samples before and after demagnetization respectively. For the cleaned declinations, the mean and standard deviation are shown by a horizontal bar.

on which we measured only declination. Fig. 3 shows that relatively few samples have high inclinations.

Table 2 gives the Fisher statistics and virtual poles for those normal and reversed samples which have directions within 40° of the axial dipole field. For direct comparison we have inverted the reversed directions. Note that the two polarities are not significantly different, while both differ from the axial dipole field in the sense of low inclination and easterly declination. Such a departure from the model of an axial, geocentric dipole for the time-averaged geomagnetic field was first recognized by Wilson (1970) and has been reconfirmed by later analyses of all palaeomagnetic data (Wilson & McElhinny 1974). According to Wilson, late Cenozoic virtual poles are 'far-sided and right-handed', meaning that the longitude of the virtual pole relative to the site (the last column in Table 2) is between 90 and 180°.

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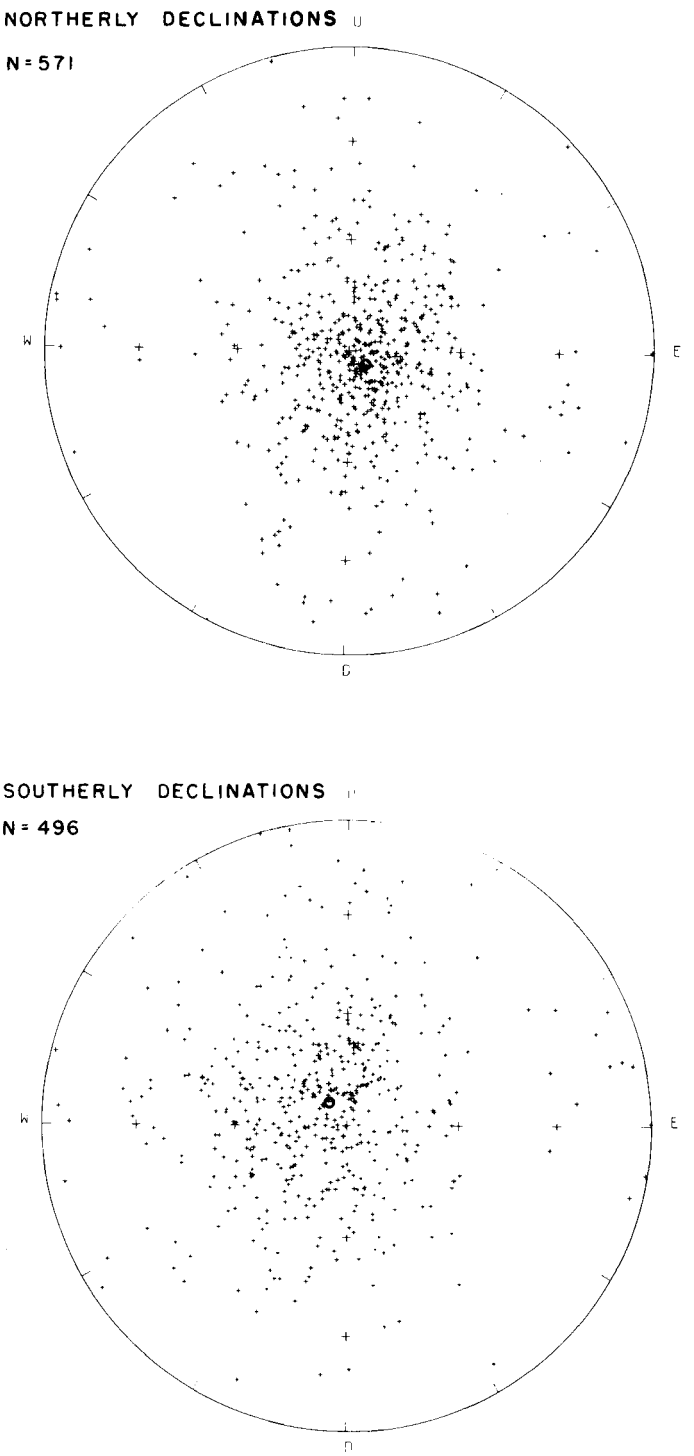


Figure 3. Stereographic plot (Schmidt equal-area net) of remanent directions after partial AF demagnetization with projection on to the plane $D = \pm 90°$. The two hemispheres correspond to northerly and southerly declination. The small circles denote the Fisher mean direction for all samples within 40° of the axial dipole field (Table 2).

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Table 2. Fisher mean directions and virtual pole positions.

	N	k	α_{95} (°)	D (°)	I (°)	θ_p (°)	$\Delta\lambda$ (°)
Normal	466	16.7	1.6	3.9	4.4	4.8	126
Reverse	382	13.2	2.0	6.0	7.0	6.2	104
Total	848	14.9	1.3	4.9	5.6	5.4	114

As described in the preceding section our procedure leading to a single cleaned direction for each sample did not take into account the angular difference between different samples at a site so long as the polarity agreed. Likewise our interpretation of magnetic zones (described below) did not use the intrasite angular difference. Nevertheless it is certainly of interest to examine retrospectively the intrasite angular dispersion. To do this we first computed the ratio R/N at each site with several vector measurements. (N is the number of vector measurements, R is the length of the sum of unit vectors.) The R/N was converted to angle by taking the inverse cosine. The resulting intrasite angular deviations are plotted as histograms in Fig. 4. For the 1973 and 1974 sites the averaged intrasite angular standard deviations are 28 and 18° respectively. We attribute the decrease to the change in field technique.

Low intrasite dispersion is commonly used as a necessary condition for palaeomagnetic results to be reliable. In this sense the deviations shown in Fig. 4 are alarmingly large. However, the critical reader should note that the histograms are for the worst rocks, not the average rocks: for 1973 samples we made only one vector measurement per site (and hence no measure of dispersion) where the polarities agreed within a site and between consecutive

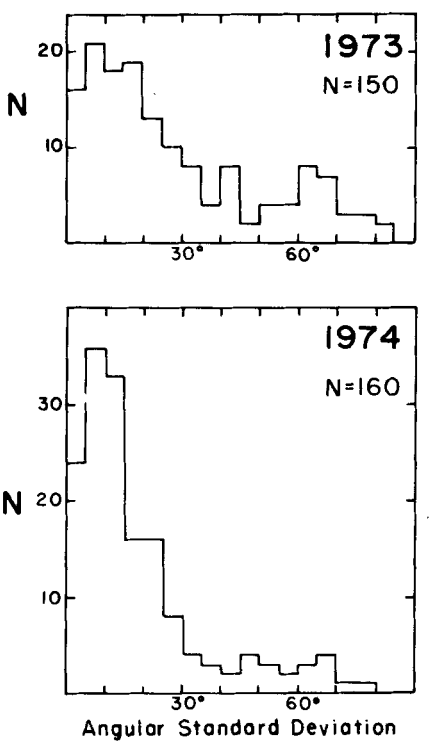


Figure 4. Histograms of intrasite angular deviations for samples taken in 1973 and 1974.

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Table 3. Fisher statistics on magnetozones of the Shungura and Usno Formations. Equal weight was given to each site.

Zone	Polarity	N	K	α_{95}	D	I
Shungura Formation						
L7/L9	*R*	9	1.4	72.1	184.2	2.6
L5/L6	*N*	9	13.8	14.3	3.4	5.2
H7/L4	*R*	102	2.0	13.8	188.2	−14.9
G26/H6	*N*	46	11.1	6.6	7.6	4.4
G24/G26	*R*	13	1.2	77.8	180.5	68.5
G23	*N*	2	–	–	10.8	9.7
G13/G23	*R*	29	1.7	30.9	192.3	−6.4
G12/G13	*N*	12	12.6	12.7	2.1	0.3
G9/G11	*R*	15	1.3	64.1	216.3	4.4
G4/G8	*N*	17	4.5	18.9	2.3	12.9
F4/G4	*R*	24	2.1	28.3	197.8	0.6
F2/F4	*N*	12	4.6	22.8	357.4	−5.5
D5/F1	*R*	55	1.9	19.9	193.2	−11.5
D4	*N*	14	2.3	34.6	359.3	−17.9
C9/D3	*R*	28	3.0	19.2	172.6	−17.7
B3/C8	*N*	128	10.5	4.0	4.0	3.2
B2	*R*	11	1.6	53.2	172.7	−28.1
B2	*N*	3	15.0	32.9	7.0	−4.3
B1/B2	*R*	7	1.4	82.9	21.8	35.5
A/B1	*N*	23	4.2	16.9	0.6	−6.6
Bas 1/A	*R*	41	1.2	43.7	204.6	−29.1
Bas 1	*N*	4	44.2	13.9	1.4	−15.5
Usno Formation						
U17/U19	*N*	8	6.5	23.5	12.4	32.9
U16	*R*	1	–	–	190.0	−37.0
U14	*N*	2	–	–	20.2	14.2
U12/U14	*R*	15	1.7	44.6	195.0	−10.3
U7/U11	*N*	26	3.8	17.0	12.9	11.4
U1/U6	*R*	12	1.3	71.7	165.7	33.5
U(Basal)	*N*	2	–	–	0.0	55.6

sites. The 1974 sampling concentrated on stratigraphic intervals where previous data were scattered.

After defining magnetozones (see below) in the Shungura and Usno Formations, we computed Fisher statistics for each zone (Table 3). For this computation each site (not each sample) was given unit weight. For sites with more than one sample on which both declination and inclination were measured, the mean direction was used in the computation. An appropriate quality measure for each zone is the Fisher Precision Index (K). Note that it is generally higher for our normal zones than for our reversed zones (Table 3). This is an effect of the occurrence of normal samples in reversed zones which are thought to be due to secondary magnetization (see below). The only zone for which the mean direction is in conflict with the assigned polarity is the zone labelled B1/B2 in Table 3. This zone is discussed in detail in the section on polarity zonation.

Stratigraphic plots of magnetic directions

We decided as before to consider separately the Shungura Formation exposures near the American Camp (Fig. 1(a)–(d)) and near Kalam (Fig. 1(e)). The reasons for this are the

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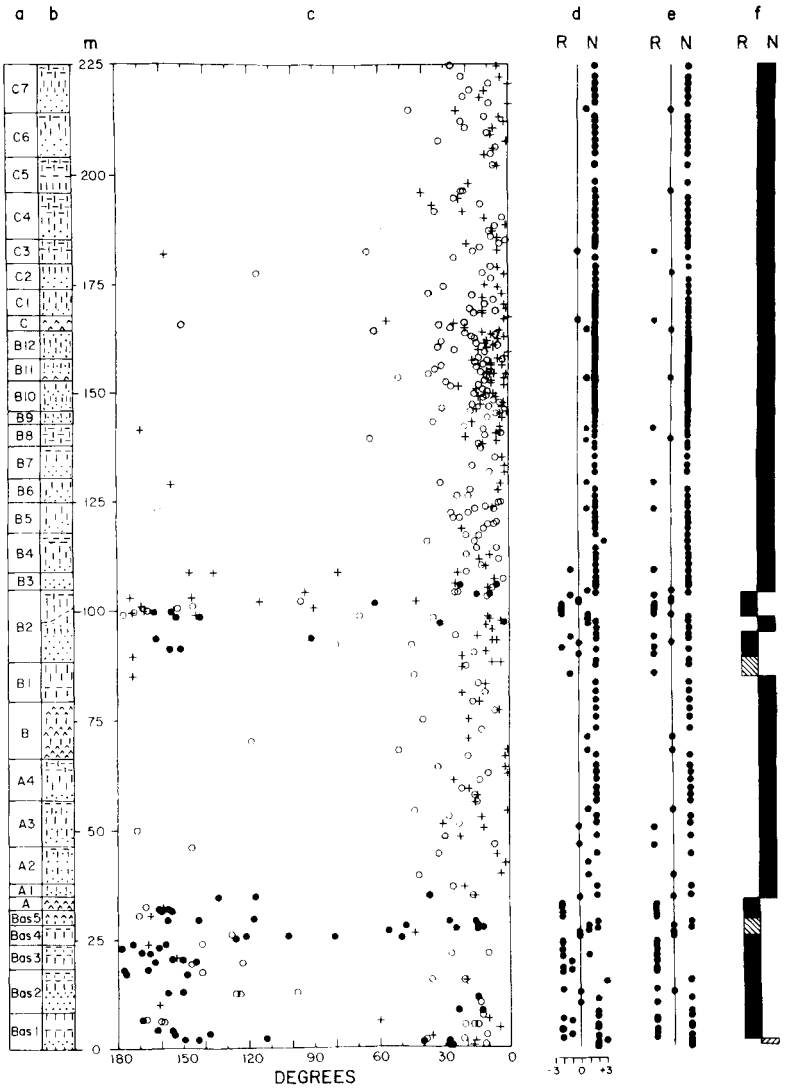


Figure 5 (a)

Figure 5. Magnetostratigraphic columns for the Shungura Formation near American Camp. Column (a) identifies lithologic members and submembers. Column (b) is simplified lithology (dots = sand; vertical dashes = silt; horizontal dashes = clay; chevrons = tuff; combinations used where appropriate). Column (m) is stratigraphic height from the base of the formation in metres. Column (c) shows the angle between the cleaned remanent direction and the axial dipole field ($D = 0°$, $I = 10°$) for each sample. Solid circles denote 1974 samples, open circles denote 1973 samples, and crosses denote those 1973 samples for which inclination was not measured. Columns (d) and (e) give the site polarities by the methods of Shuey *et al.* (1974) and Hillhouse *et al.* (1977) respectively. Column (f) is our final polarity zonation. Hachures indicate doubt.

difference in sedimentation rate between the two areas (Shuey *et al.* 1974, Table 2) and the possibility that the magnetozone boundary correlated with the top of the Gilsa/Olduvai event is lower in the lithologic section near Kalam (Shuey *et al.* 1974, Figs 3 and 4). For all samples we plotted the angle between cleaned direction and dipole field (Figs 5–7) versus stratigraphic position. For those 1973 samples without inclination measurements, this angle was calculated on the basis of declination alone.

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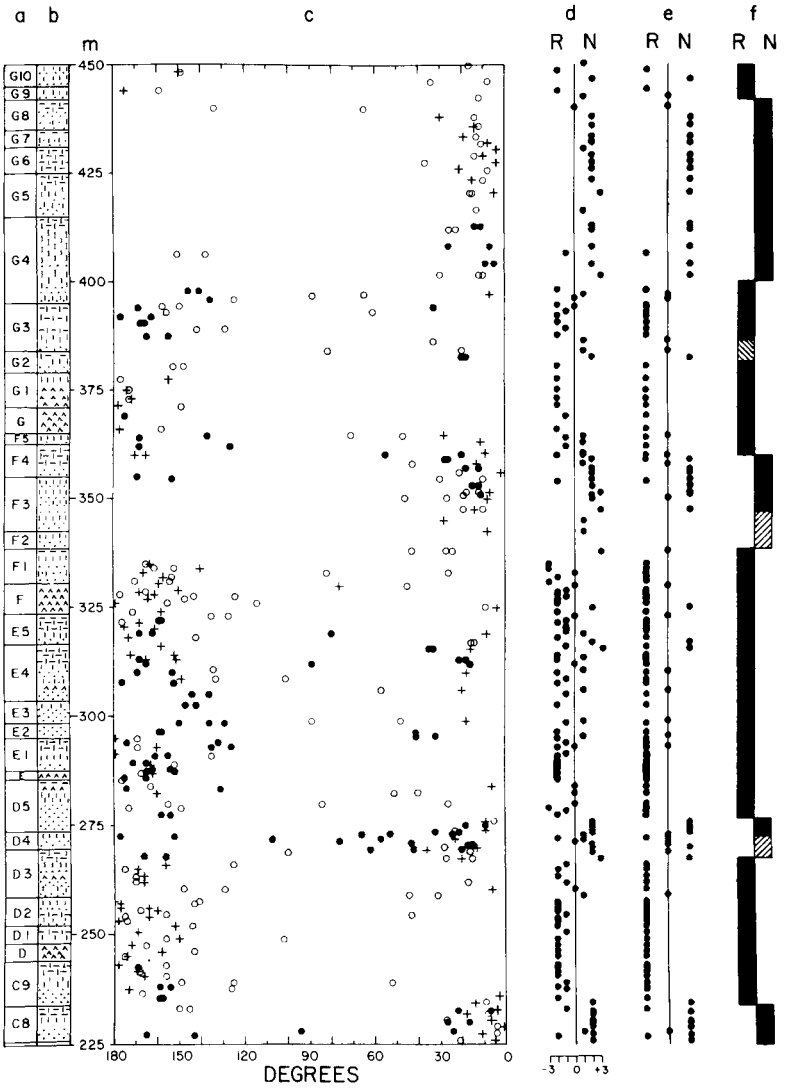


Figure 5 (b)

Statistical asymmetry between normal and reverse groups is evident in the stratigraphic plots (Figs 5–7). Portions of the section in which data are concentrated on the right-hand side of the diagram (normal samples) tend to be fairly clean, with few reversed or intermediate samples in these intervals. This is most apparent between 30 and 90 m, 105 and 225 m and 560 and 630 m on Fig. 5. In contrast, portions of the section which have a large number of data points on the left side of the diagram (reversed samples) also have a large number of normal and indeterminate samples associated with them. This is clearly seen between 2 and 30 m, 250 and 330 m and 635 and 690 m on Figs 6 and 7. The same features were noted by Hillhouse, Ndombi & Brock (1977) for a sedimentary section east of Lake Turkana (formerly Lake Rudolf).

Comparison of 1973 and 1974 collections

The newer data differ systematically from the older measurements on the same rock units in the sense of a shift towards reverse polarity. We infer that the improved sampling pro-

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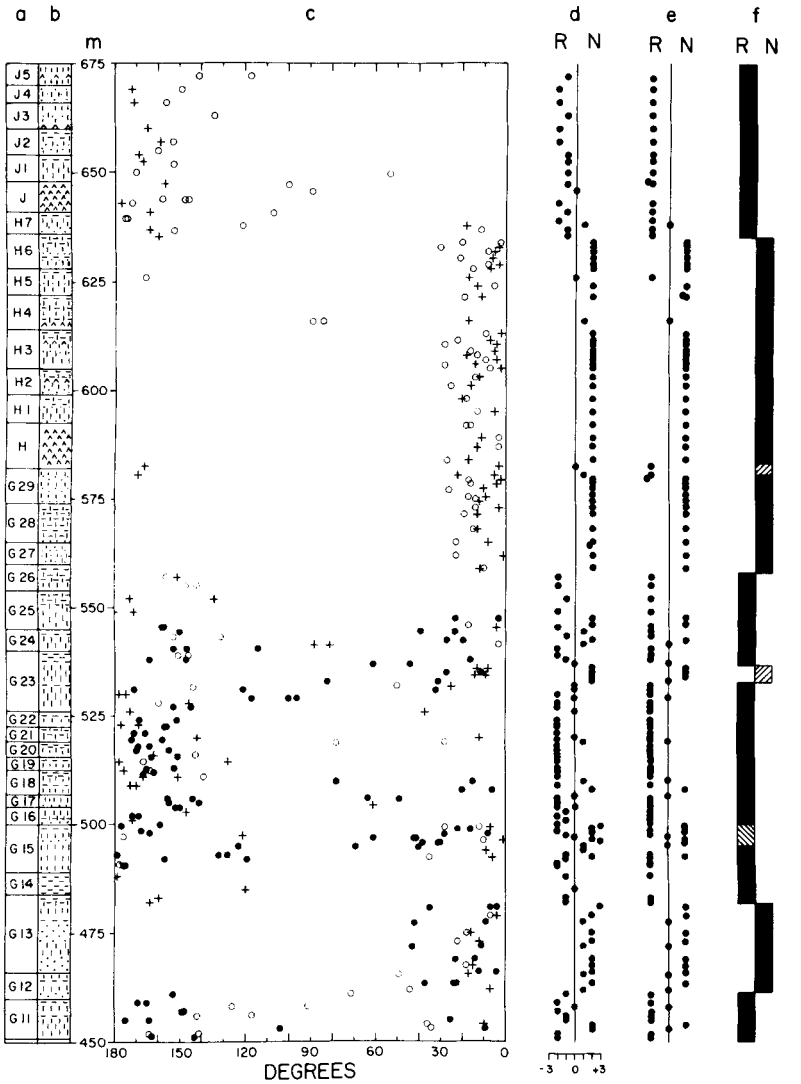


Figure 5 (c)

cedure was effective in reducing secondary magnetization along the present field. The effect of the additional data is generally to make the inference of magnetozones less speculative and more precise. Only in one respect did the resampling lead to a qualitative change in zonation: previously the polarities from submembers BAS-2 and BAS-3 had been indeterminate with some normal samples; we proposed a tentative normal zone. Most of the new directions from these rock units are of reversed polarity. The few normal 1974 samples represent the basal sand of BAS-2. This new dominance of reverse polarity in submembers BAS-2 and BAS-3 makes a qualitative change in our proposed polarity chronostratigraphy. Previously the reversed rock at the top of the Basal Member had been identified with the Mammoth event; now we propose to correlate with the top of the Gilbert Epoch. A complete reanalysis of the polarity chronostratigraphy will be presented below.

A similar difference between 1973 and 1974 measurements is found in the middle of Member E. The 1973 samples were sparsely distributed with a high proportion of sands. The

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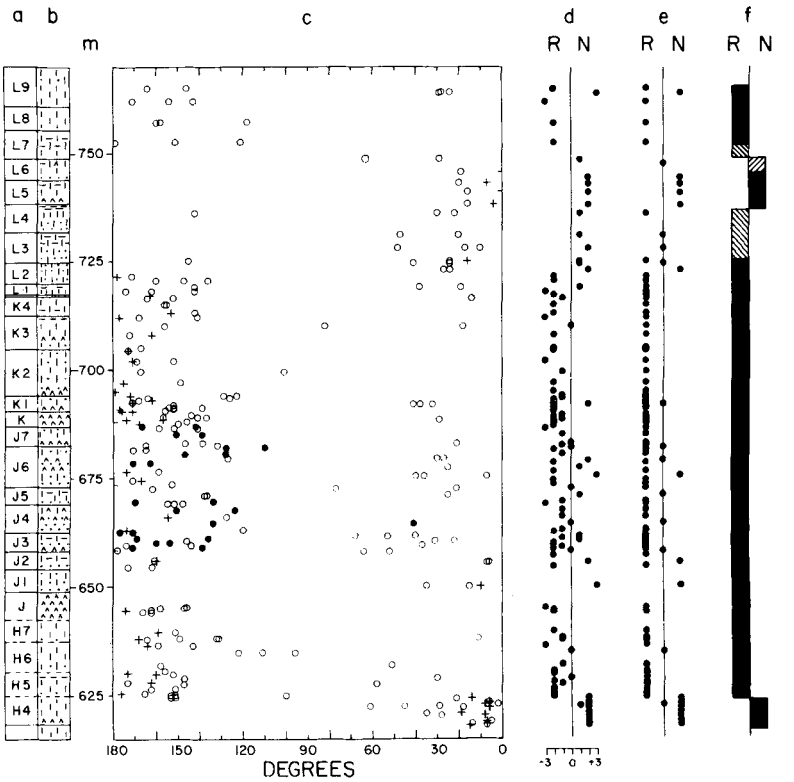


Figure 6. Magnetostratigraphic columns for the Shungura Formation near Kalam. Column headings are explained in the caption to Fig. 5.

1973 measurements gave normal and indeterminate polarities. The 1974 samples were predominately reversed with all normal samples being sand. However, in this case no change in our magnetic zonation resulted. Previously (Shuey *et al.* 1974) we chose to discount all measurements from middle Member E and postulate reversed polarity on the basis of interpolation between the more consistently reversed measurements in upper and lower Member E. This is now supported by more substantial data.

Polarity zonation

In order to reduce further the directional data each sample was declared to be normal or reversed if the calculated angle from the dipole was in the range 0–40° or 140–180° respectively. Then the several samples from the same site were reconciled according to two alternative schemes. The first is the 'algebraic sum' method we used previously (Shuey *et al.* 1974), in which each sample is assigned a value of + 1 if it is of normal polarity, – 1 if it is of reversed polarity and 0 if the sample is indeterminate. All values from each site are added. The method of Hillhouse *et al.* (1977) assigns polarities to each level with a bias towards reversed polarity. Their rules (slightly modified) are: (1) if a set of samples from a site contains at least one reversed sample, the site is considered reversed; (2) if a set of samples from a site contains normal and indeterminate samples only, the level is considered indeterminate; (3) if a set of samples from a site contains only normal samples, it is considered normal. Each scheme yields a polarity log, shown in columns (d) and (e) of Figs 5–7. Finally we interpreted the magnetozones shown in column (f), Figs 5–7. Admittedly the

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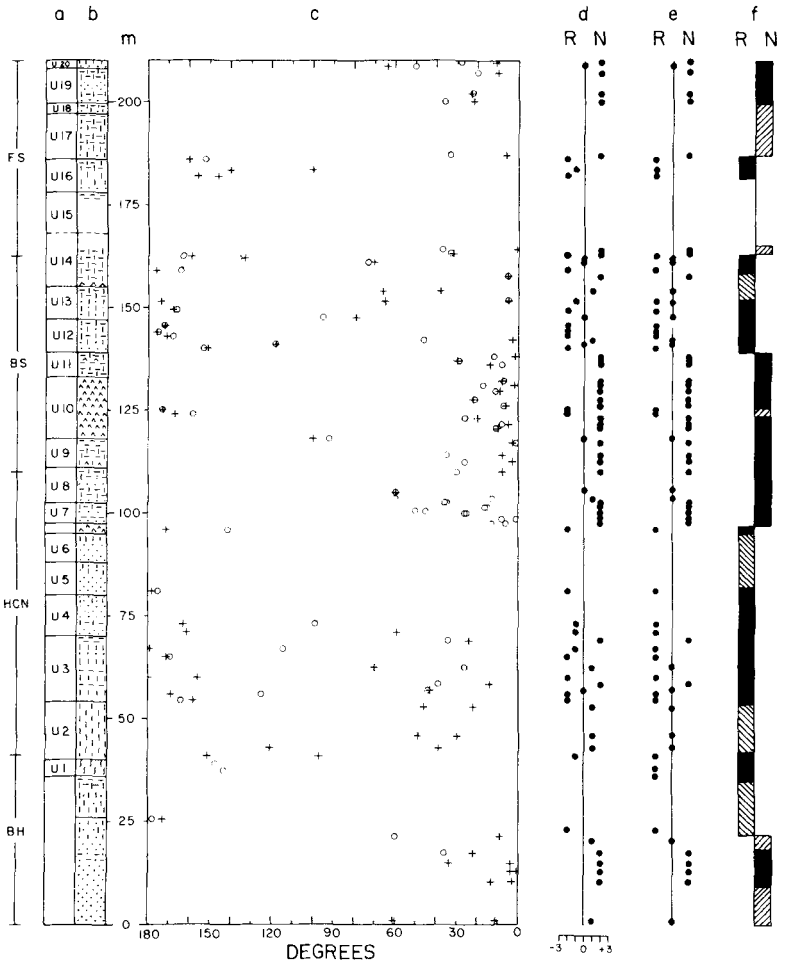


Figure 7. Magnetostratigraphic columns for the Usno Formation. Column headings are explained in the caption to Fig. 5. The locality symbols to the left of column (a) refer to Fig. 1(f).

interpretation is still somewhat subjective in parts of the section, and the reader might be inclined to draw column (f) somewhat differently. A few of our interpretations warrant additional comment as follows.

In 1973 we had sampled the Basal Member of the Shungura Formation entirely at one hill designated P748. Submembers BAS-2 and BAS-3 were indeterminate or questionably normal while submembers BAS-4 and BAS-5 were reversed. In 1974 we resampled submembers BAS-1, -2 and -5 at P748 and submembers BAS-3 and BAS-4 at an adjacent hill designated P64 where we had previously sampled the normally magnetized Member A (Fig. 1(d)). The new data showed BAS-2 and BAS-3 to be reversed but BAS-4 to be normal. Regrettably we did not sample BAS-5 at location P64 in 1974. Instead we resampled at P748 and found it still reversed. This presents an ambiguity of interpretation which cannot be resolved without a return to the Omo. One possibility is that the reversed magnetozone which extends to the top of P748 terminates at the boundary between BAS-3 and BAS-4 at P64. This interpretation predicts that BAS-5 would be normal at P64 even though we have twice measured it as reversed at P748. It also suggests that the two hills may have been miscorrelated which does not appear likely, especially since the lower part of Tuff A has been

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sampled twice at P64 and has been found to be reversed both times. An alternative interpretation is that the normal magnetization found for BAS-4 at P64 is secondary. This interpretation is favoured in column (f), Fig. 5.

In 1973 only a few reversed samples were found in submember B-1, but they were ignored in the interpretation. More turned up in lower B-2 upon resampling, but still a bias towards reversed polarity is necessary in order to postulate a reversed rather than an indeterminate zone (*cf.* zone B1/B2 in Table 3). We show a reversed zone in Fig. 5 column (f) partly because of the consequent good fit to the Mammoth event (Fig. 8). It might also be mentioned that three of the samples from lower B-2 which are shown as normal in Fig. 5, column (c), were actually reversed before AF demagnetization.

Our previous interpretation showed a normal zone from upper D-3 to lower D-5. The new samples from a slightly different location (Fig. 1(c)), confirmed normal polarity only for the upper part of the zone; the lower part now appears reversed. The sands in the submembers D-3 to D-5 all exhibited such severe VRM that we could not assign any remanent direction to them. Thus there is a possibility that still further sampling and laboratory analysis might entirely eliminate the normal zone still shown.

The 1973 samples showed a normal zone in upper G-12 and lower G-13. It was considered questionable (Shuey *et al.* 1974, Fig. 3) because most of the samples were from concreted sand. However, the 1974 samples taken 4 km away (Fig. 1(b) and (d)), again show normal polarity, so the evidence for a normal zone is strongly reinforced. In the rest of the interval from G-9 to G-23 the previous interpretation of reversed polarity is substantiated by the 1974 data (Fig. 5). We still show a very thin normal subzone within G-23 because both 1973 and 1974 samples were consistently normal. However, as stated before the associated time interval would amount to only a few thousand years at the most. We omit this possible subzone from Fig. 8.

Correlation between the Shungura and Usno Formations

De Heinzelin *et al.* (1976) have independently correlated the sections of the Usno and Shungura Formations, primarily on the basis of lithology. In particular, Shungura Tuff A and Tuff B are correlated with the Usno tuffs between U-6 and U-7 and in U-10 respectively. This is entirely consistent with the magnetozones, which in each formation show a reverse to normal transition at the lower tuff and normal polarity continuing through the upper tuff (Figs 5 and 7). We would further identify the predominantly reversed part of the section in Shungura submembers B-1 and B-2 with the predominantly reversed part of the Usno Formation from U-11 to U-16. In the Shungura Formation this zone is relatively thin, which may be attributed to an erosional contact within submember B-2 (see below).

Magnetochronology

Potassium–argon dates (Table 4) consistent with stratigraphic order are available for several tuffs and a lava from the Shungura and Usno Formations (Brown & Nash 1976). As a preliminary step in chronologic interpretation of the magnetozones, we placed the rock units on a time scale solely on the basis of the potassium–argon dates (Fig. 8, columns (a) and (c)). Each midpoint age was taken as correct, and the other strata were placed on the timescale by assuming constant sedimentation rate between consecutive dated levels and extrapolating beyond the highest and lowest dated levels. Notice in Fig. 8 column (a) that the dates on the Usno Formation have relatively large uncertainties. The age for the members of the Shungura Formation from the Basal Member to Member C is rather tenuous because there is just one date in B-10.

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Table 4. Potassium–argon dates for Shungura and Usno Formations.*

Dated horizon	Age (Myr)
Tuff L	1.34 ± 0.10
Tuff H-2	1.83 ± 0.06
Tuff G	1.93 ± 0.10
Tuff F	2.03 ± 0.08
Tuff D-3	2.12 ± 0.11
Tuff D	2.39 ± 0.08
Tuff B-10	2.95 ± 0.07
Tuff U-10	2.97 ± 0.30
U-1 (basalt)	3.31 ± 0.23

*Data taken from Brown & Nash (1976) and reduced by averaging multiple runs on the same horizon.

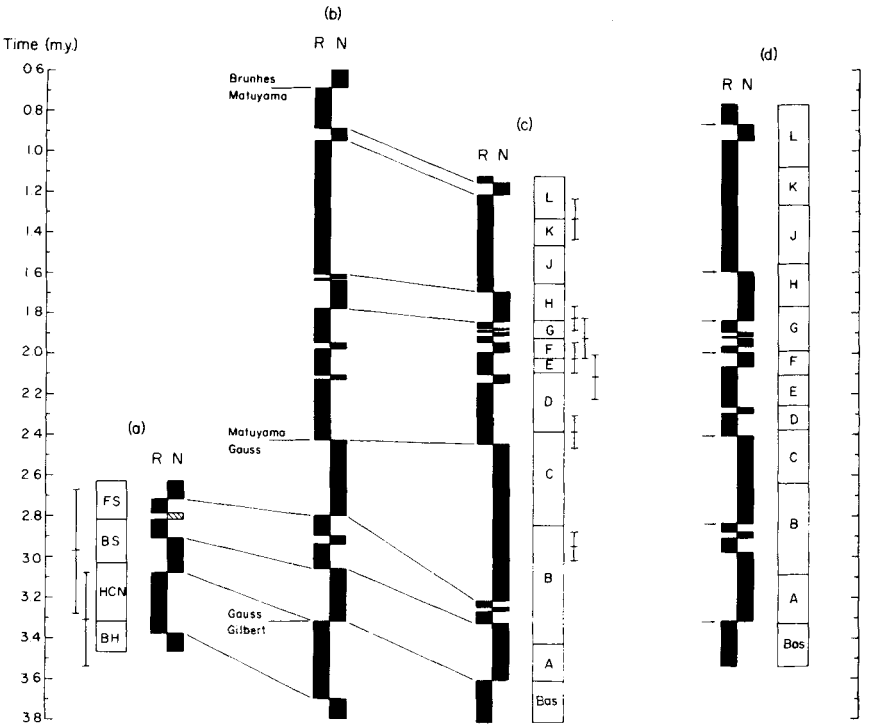


Figure 8. Magnetochronology of the Shungura and Usno Formations. Column (b) is the standard polarity scale of Cox (1969). In columns (a) and (c) the Usno and Shungura polarity zones from Figs 5–7 are placed on the time axis using radiometric dates (Table 4). The dates and their standard deviations are shown beside each column. The error bars incorporate the scatter in mean dates as well as the standard deviation for individual determinations. Constant sedimentation rate is assumed in order to interpolate between dates and extrapolate beyond them. The Camp and Kalam sections (Figs 5 and 6 respectively) are joined at submember J4. Finally in column (d) the Shungura polarity zones are placed on the time axis by assigning dates to the prominent zone boundaries (Table 5) indicated by arrows. Again constant sedimentation rate is assumed to interpolate the times of member boundaries.

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We then compared our scale with the standard polarity scale of Cox (1969), reproduced in Fig. 8, column (b). As before, the identification in the Shungura magnetozones of the Matuyama/Gauss boundary and the Gilsa/Olduvai event (in Member H) seemed quite unambiguous. We again correlated the normal zone in Member L with the Jaramillo event, and the reversed zone from B-2-5 to B-2-8 with the Kaena event. This interpretation implies that the potassium–argon dates on B-10 and Tuff L are a little too old. Actually this need not be true for the latter since alternatively one could postulate an unusual amount of unrecorded time in lower Member L.

As explained previously, the addition of the 1974 samples made enough change in the zonation of the lower Shungura Formation that we now propose to correlate the reversed zone in the Basal Member with the top of the Gilbert Epoch instead of the Mammoth event. The potassium–argon data are loose enough to allow either interpretation. Consideration of sedimentation rates definitely favours the new interpretation, as discussed below.

There is considerable uncertainty about the number of normal events between the base of the Gilsa/Olduvai event and the Matuyama/Gauss boundary, a time range roughly from 1.8 to 2.4 Myr. Deep-sea cores frequently record no events at all (Opdyke 1972). Sea-floor magnetic anomalies have been interpreted with one (Heirtzler *et al.* 1968), two (Emilia & Heinrichs 1972) or three (Rea & Blakeley 1975) events. An early Pliocene section in New Zealand shows three events (Kennett, Watkins & Vella 1971; Kennett & Watkins 1974). A recent compilation of potassium–argon dates on lavas of known polarity (Dalrymple 1972) would imply three or four normal events if the standard deviations on the dates were accepted as absolute errors. For the Shungura section we have four or possibly even five zones which could be interpreted as short normal intervals in the lower Matuyama Reversed Epoch. Three of these (F-2 to F-4, G-4 to G-8 and G-13) have been sampled twice in at least slightly different locations, and are fairly sharply defined. The other two (in D-4 and G-23-2) were found to be of smaller stratigraphic extent after resampling and reanalysis. If additional palaeomagnetic work in the Omo becomes possible these zones should be investigated in greater detail.

Sedimentation rates

Having used the potassium–argon dates to identify our Shungura magnetozones with the standard magnetochronologic intervals, we then placed the rock units on a timescale using our own selection from the body of literature on the absolute times of the interval

Table 5. Depositional rates computed between magnetic boundaries.

Stratigraphic level	Magnetic boundary	Age assigned to boundary (Myr)	Rate of deposition of intervening sediments (cm/1000 yr)
Tuff A	Top of Gilbert epoch	3.32	
			15
B-2/B-3 contact	Top of Kaena event	2.84	
			30
C-8/C-9 contact	Top of Gauss epoch	2.41	
			31
F-4/F-5 contact	Top of Reunion event	2.00	
			124
G-26/G-27 contact	Bottom of Gilsa/Olduvai event	1.84	
			32
H-6/H-7 contact	Top of Gilsa/Oluvai event	1.60	
			16
L-6/L-7 contact	Top of Jaramillo event	0.88	

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boundaries. For seven judiciously-spaced zone boundaries, marked by arrows in Fig. 8, column (d), we assigned the ages given in Table 5. Then the other rocks were placed on the timescale by assuming constant sedimentation rates between consecutive dated zone boundaries. Notice in Table 3 the relatively high rate computed for Member G. This is attributed to rapid sedimentation for the lacustrine facies in submembers G-14 to G-26. Notice also that the rate of sedimentation decreases at both the top and bottom of the section.

The thickness of Shungura Formation sediments associated with the time interval between Mammoth and Kaena events was a major consideration in our revision of the magnetochronology. Previously the 75 m of section between Tuff A and the erosional contact in B-2 had been ascribed to the Mammoth–Kaena gap. If the gap has a time value of 0.05 Myr, the depositional rate would be 150 cm/1000 yr. This was anomalous because otherwise a rate this large was found only in the lacustrine facies of Member G. The lithology of Member B indicates a fluvial environment of deposition, for which rates of 20 to 30 cm/1000 yr seem the norm (Table 5).

Previously we remarked that the thickness of Shungura Formation sediments associated with the Kaena event was anomalously low. With our revised identification of the Mammoth event, the entire submember B-2 is characterized by low rates of sedimentation. These range from 6 to 10 cm/1000 yr, depending on where the zone boundaries are picked and how much time is assigned to the Mammoth and Kaena events and the gap between them. This is not completely unreasonable, because there are a number of diastems within submember B-2 on the basis of which this submember has been divided into eight units. The most evident of these is an erosional contact between B-2-4 and B-2-5. In some exposures both B-2-5 and B-2-6 are absent and B-2-7 lies directly on B-2-4. Both the diastems and loss of section indicated by the absence of B-2-5 and B-2-6 in some outcrops might cause the low computed sedimentation rate for this stratigraphic interval.

In constructing column (d) of Fig. 8 we wanted to fix in time some level immediately below the lacustrine section which extends from G-14 to G-26, so that the relatively high sedimentation rate we have in the lower Matuyama Epoch would be associated more closely with the lacustrine facies. As before we chose to match the normal zone from F-2 to F-4 with the event convincingly documented on Reunion Island by McDougall & Watkins (1973). This gives the sensible sedimentation rates shown in Table 5. There is another possible way to match the Shungura data with other magnetochronologic information, and this is suggested by column (c) of Fig. 8. The potassium–argon date on D-3 would be more closely honoured if the normal zone in D-4 were identified with the event at 2.1 Myr shown by Cox (1969). The latter interpretation implies that the rate of accumulation of sediment for lower Member D was low, about 8 cm/1000 yr. This is comparable with the rate computed for B-2, and is perhaps not unreasonable.

Status of Omo chronology

Aside from any inferences one might wish to draw about the lower Matuyama Epoch, the principal importance of the magnetostratigraphy is to establish time lines in the Shungura Formation against which rates of evolution, changes in climatic indicators, time ranges of fossil taxa and changes in sedimentation can be viewed. The time lines between the Shungura and Usno Formations also allow consideration of isochronal variation in fauna and environment.

The identification of the zone boundary between Shungura C-8 and C-9 with the Matuyama/Gauss boundary is so strongly supported by the potassium–argon dates that we

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regard it as indisputable. Likewise we have no reason to doubt that the long normal interval from G-27 to upper Member H, including a tuff dated at 1.83 Myr represents the Olduvai event of deep-sea cores and magnetic lineations. We have previously mentioned (Shuey *et al.* 1974) the similarity of the fauna in this magnetozone to that at the type locality (Olduvai Gorge in Tanzania) and have pursued the Omo biostratigraphy in a separate paper (Shuey *et al.* 1978).

If the normal magnetization in middle Member L is not secondary, we must identify this zone with the Jaramillo event for the reasons previously given (Shuey *et al.* 1974). An alternative hypothesis is that the primary polarity is reversed throughout Member L and the top of the Shungura Formation is 1.2 instead of 0.8 Myr old. To test this we did sample Member L in 1974 at a different location than that used in 1973 (see Fig. 1). These samples were not plotted on Fig. 6 because the final lithostratigraphic correlation with the previously sampled section is not yet available to us. However, the data do show a transition from reversed to normal and back to reversed, in support of our interpretation.

In this paper we have tentatively revised the chronology of the lower part of the Shungura Formation. However, this affects only the section below submember B-2, and it should be noted that these strata contain only 1 per cent of all specimens and only 6 per cent of all taxa are represented. Therefore the revision we proposed has essentially no effect on analyses of faunal similarity between the Shungura Formation, and other formations in East Africa (Shuey *et al.* 1978). However, the interpretation of the timing of geologic events in the lower Omo basin is definitely modified.

Acknowledgments

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